

Cyclistic Case Study

Beatriz Moraes



Marketing Objective

Our marketing objective is to drive annual membership growth by converting casual riders via data-driven campaigns.

Dataset

The datasets used were provided by Motivate International Inc. under the license: *<https://divvybikes.com/data-license-agreement>* and are suitable for the purposes of this case study.



Executive summary

Key Insights

- Spring and summer mark the peak season for casual users
- Casual users show a recreational profile, often associated with longer trips
- Members maintain steady usage during winter, reinforcing the value of the subscription plan
- Tourist and academic zones present strong conversion potential

Key Recommendation

Launch a seasonal 7-day free trial campaign targeting active casual users, supported by personalized notifications. Leverage in-app gamification and engagement through local events to boost retention and conversion.

Next Steps

- Conduct A/B testing in two pilot regions (Northeast and South)
- Deploy a real-time conversion dashboard



Methodology

Data Cleaning & Preprocessing

- Handling of null values and duplicates
- Distance calculation using the Haversine formula

Key Variables Analyzed

- member_casual, day of the week, season, time of day, ride duration, distance, station (located in Chicago), geographic zone.

Tools Used

- BigQuery (SQL) for data extraction and analysis and RStudio (R) for visualization development.



Analysis

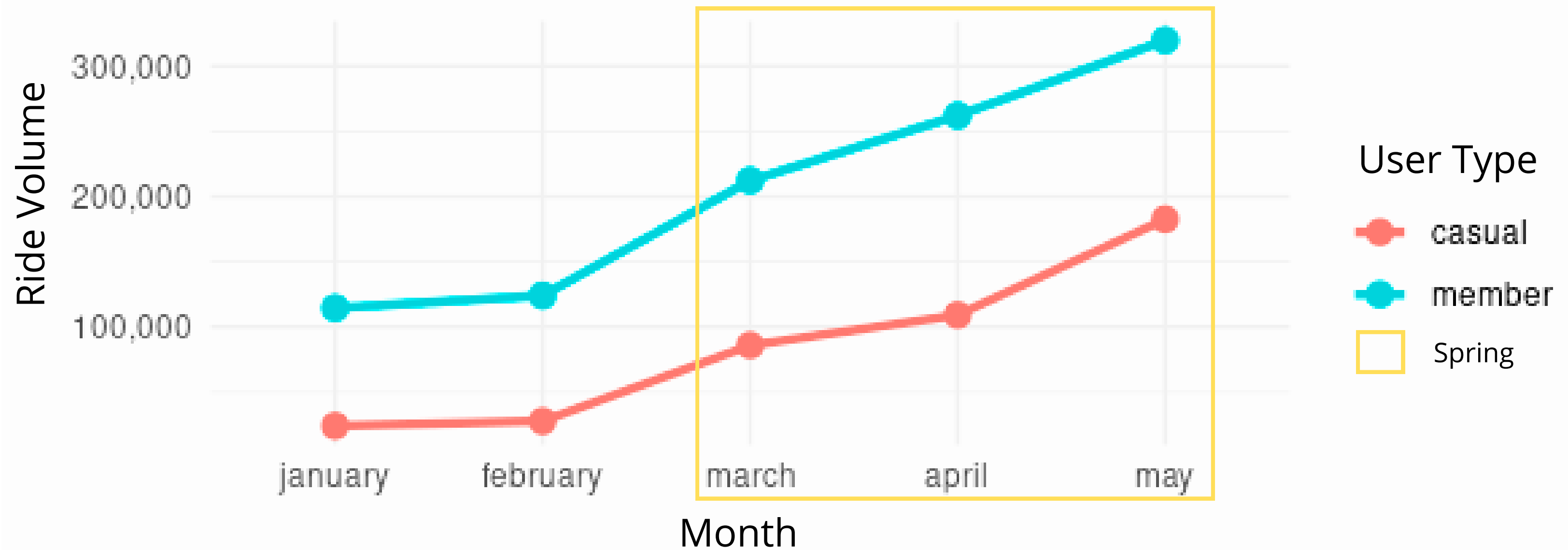
Usage Patterns: Casual vs. Member Riders

1.1 Seasonal and Time-of-Day Behavior

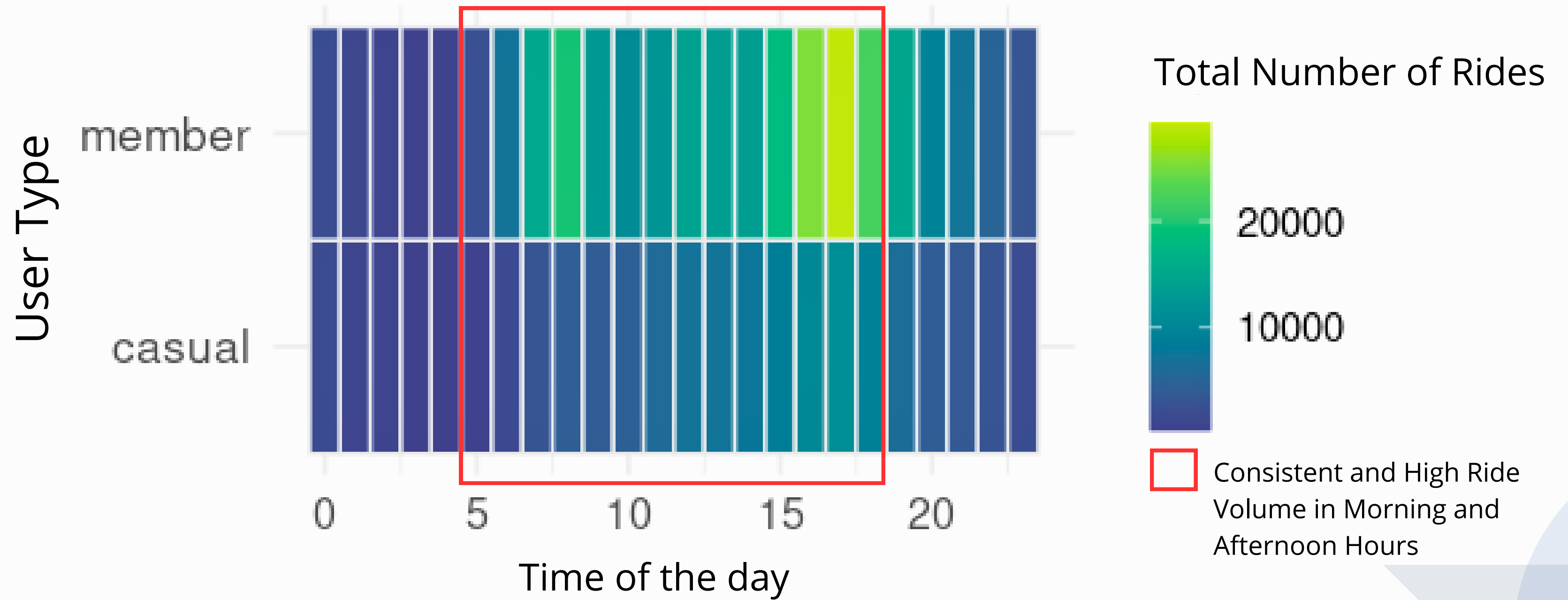
- Spring (March to May): Significant increase in casual riders; member ride volume also rises
- April and May: Consistently high usage by both groups during morning and afternoon hours



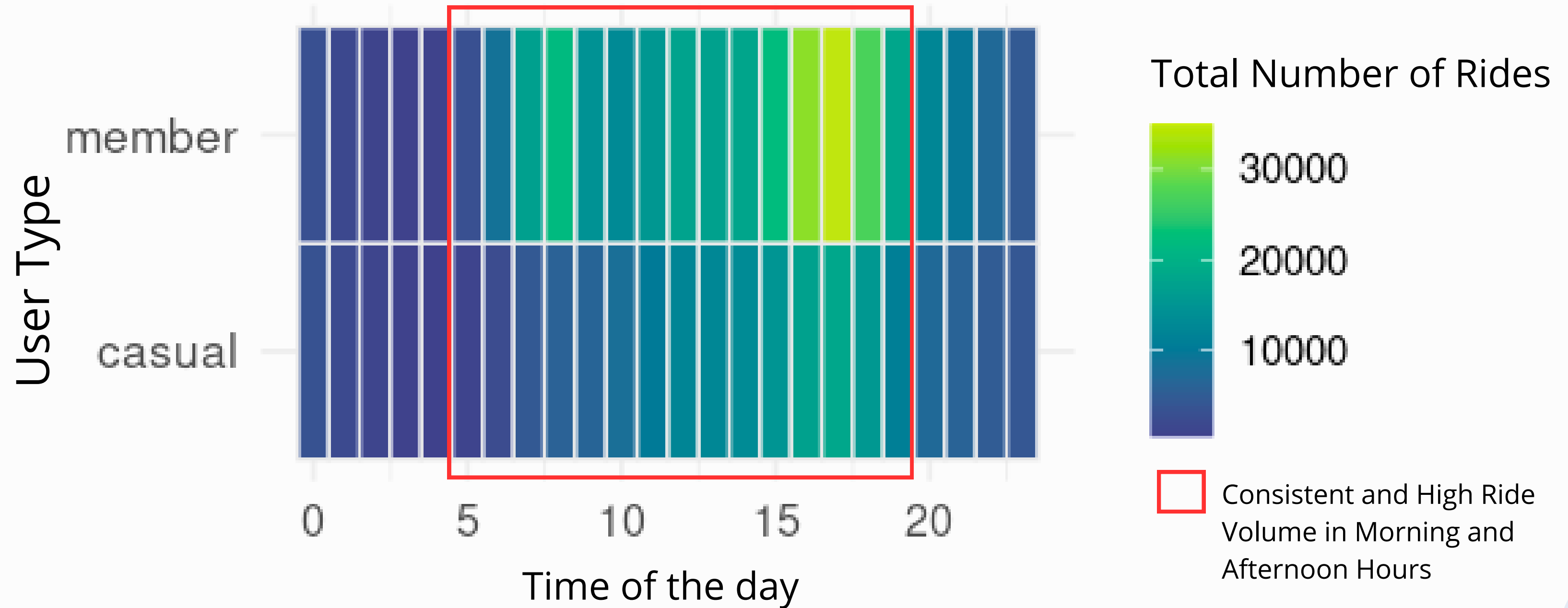
Monthly Ride Trends: Casual vs. Member Users



Ride Trends: April



Ride Trends: May



Analysis

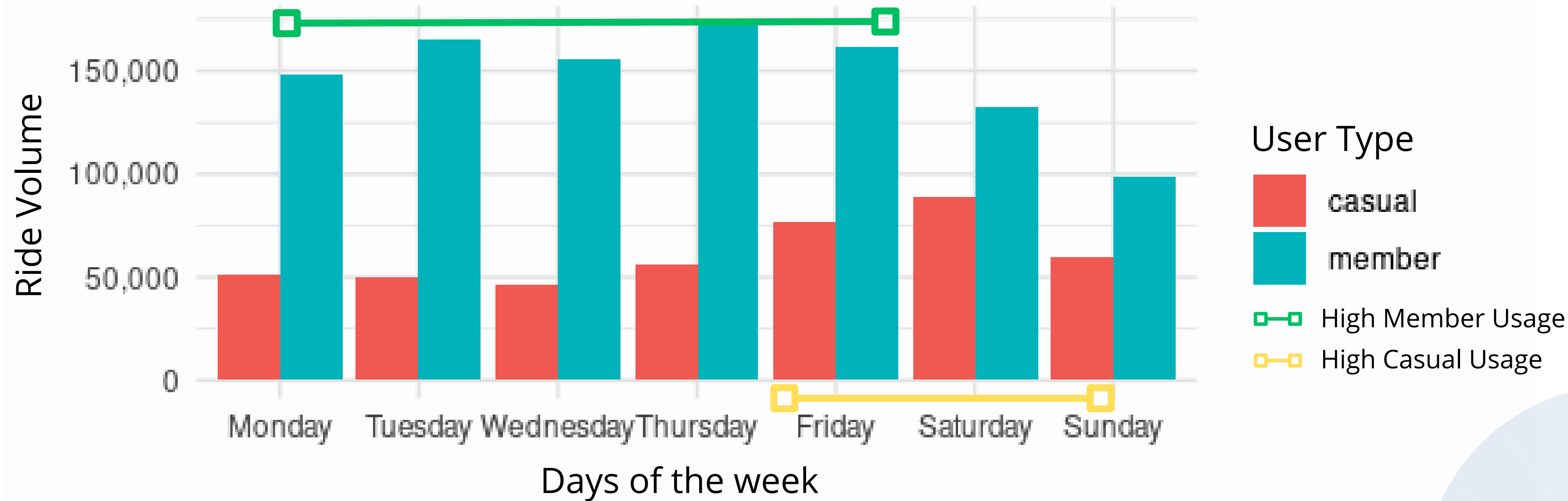
Usage Patterns: Casual vs. Member Users

1.2 Frequency by Day of the Week

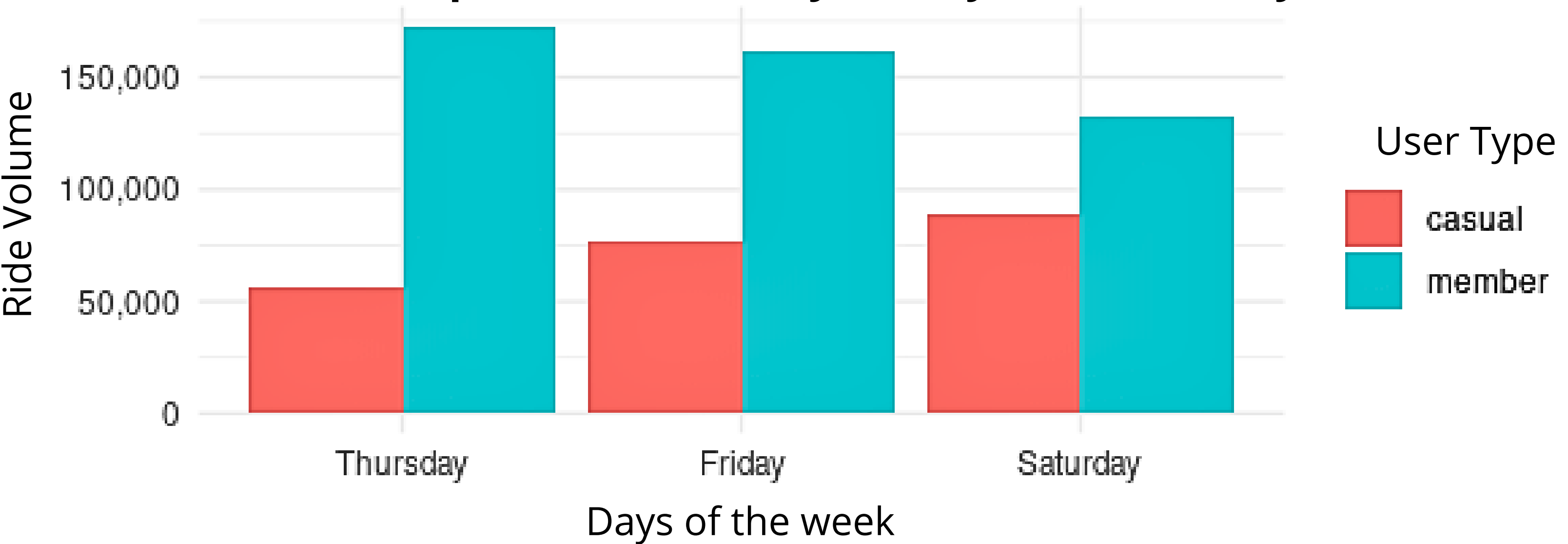
- Casuals are concentrated on weekends.
- Members predominate during the week.



Usage Patterns by day of the week



Ride Comparison: Thursday, Friday, and Saturday



Analysis

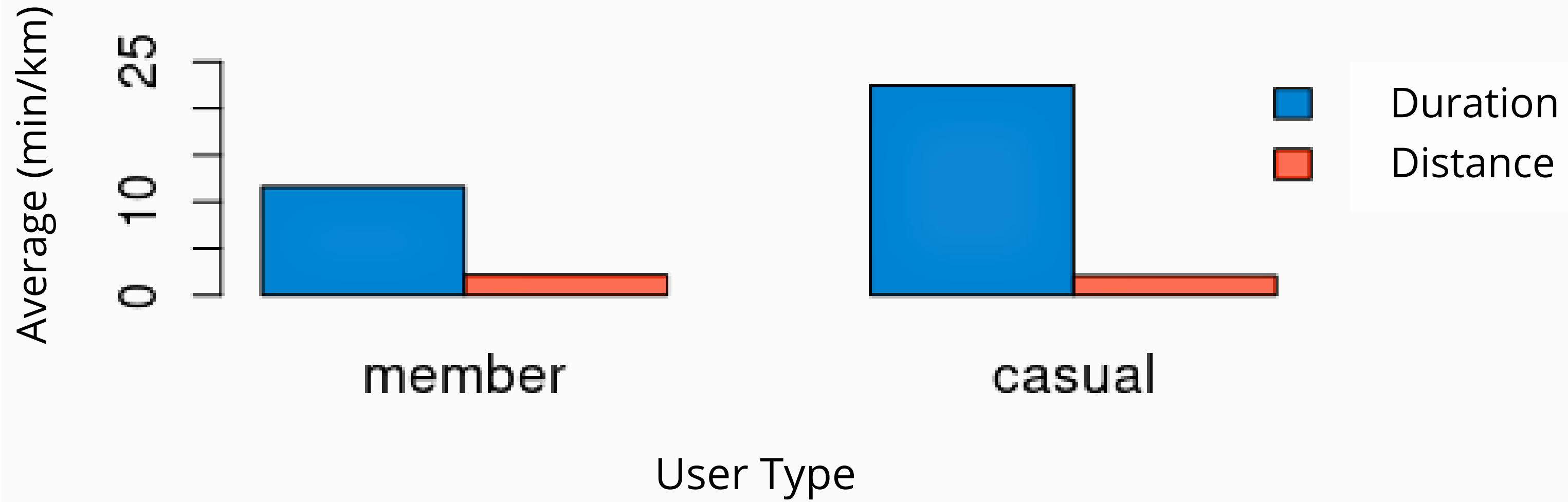
Usage Patterns: Casual vs. Member Users

1.3 Ride Profile (Duration × Distance)

- Average Duration: Members (9–11 min) vs. Casuals (20–23 min)
- Average Distance: Both ~2 km during peak months



Ride Profile



Data from March, April, and May 2025.

Analysis

Usage Patterns: Casual vs. Member Users

1.4 Usage Zones

- Top 10 stations most used by Members vs. Casual users:
 - Members: Academic zone (South), Commercial and Transport zones (West)
 - Casual Users: Tourist zone (Northeast)

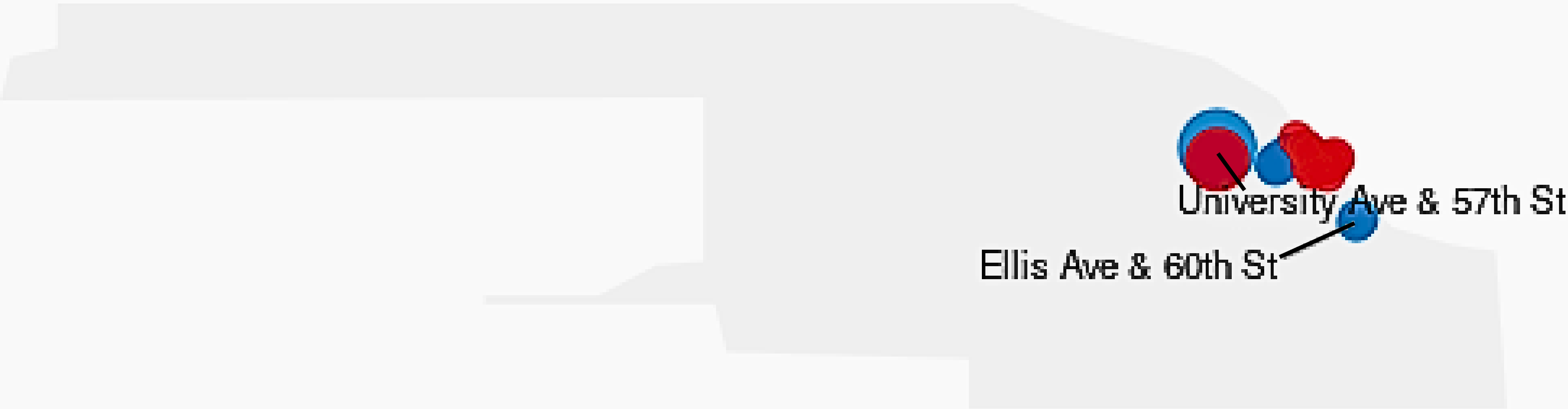


Ranking	User Type	Station	Neighborhood/Region	Main urban area
1	Member	N Paulina St & W Pierce Ave	Wicker Park / West Town	Tourist / Residential
2	Member	Kingsbury St & Kinzie St	River North / Fulton Market	Tourist / Commercial
3	Member	Clinton St & Washington Blvd	West Loop / Fulton River	Transport / Commercial
4	Member	University Ave & 57th St	Hyde Park	Academic
5	Member	Clinton St & Madison St	West Loop	Transport / Commercial
6	Member	Canal St & Madison St	West Loop / Greektown	Transport / Commercial
7	Member	Clark St & Elm St	Near North / Gold Coast	Tourist / Residential
8	Member	Ellis Ave & 60th St	Hyde Park / Woodlawn	Academic / Residential
9	Member	Clinton St & Jackson Blvd	West Loop	Transport / Commercial
10	Member	Wells St & Elm St	Old Town / Near North Side	Tourist / Residential

Ranking	User Type	Station	Neighborhood/Region	Main urban area
1	Casual	W Hubbard St & N Paulina St	West Town / Wicker Park	Residential / Commercial
2	Casual	Streeter Dr & Grand Ave	Streeterville / Near North Side	Tourist / Commercial
3	Casual	DuSable Lake Shore Dr & Monroe St	Grant Park / The Loop	Tourist / Institutional
4	Casual	Millennium Park	The Loop	Tourist / Cultural
5	Casual	Shedd Aquarium	Museum Campus / South Loop	Tourist / Institutional
6	Casual	Michigan Ave & Oak St	Gold Coast / Magnificent Mile	Tourist / Commercial
7	Casual	Dusable Harbor	Grant Park / Downtown	Tourist / Recreational
8	Casual	DuSable Lake Shore Dr & North Blvd	Lincoln Park / North Side	Residential / Recreational
9	Casual	Theater on the Lake	Lincoln Park / Fullerton Ave	Cultural / Recreational
10	Casual	Indiana Ave & Roosevelt Rd	South Loop / Museum Campus	Institutional / Residential

Chicago's top 10 stations by user category

Data from March, April, and May 2025.



User Type ● Member ● Casual

Marketing Action Recommendations

Seasonal and Targeted Offers

- Spring/Summer: Membership discount for active casual users in April and May.
- Winter: Promotional package with reduced pricing and added benefits

Personalized Communication

- Notification for casual users who take ≥ 3 trips/month, offering a free 7-day membership trial.
- Monthly reports showing accumulated savings if they had been members.

Geographic Incentives

- Partnerships at tourist hotspots (cafés, museums) offering discounts to new members.
 - On-campus activations with quick talks highlighting membership benefits
- 

Marketing Action Recommendations

Gamification and Engagement

- Monthly challenges: Complete X rides to earn a temporary upgrade.
- “Try the Member” events at DuSable, Streeter, and Shedd stations, with giveaways and photo ops.

Monitoring and Optimization

- A/B testing of messages, creatives, and offers.
- Real-time dashboard to track conversion rates and adjust the campaign in biweekly cycles.



Analysis Limitations

The analysis was conducted using data from January to May due to technical limitations in processing the files from June and July. While this may restrict the observation of full seasonal patterns, the first five months of the year provide a solid foundation for identifying casual rider profiles and proposing strategic conversion initiatives. It is recommended that future analyses incorporate the subsequent months to validate and refine the recommendations presented.



.....

Agosto 2025

Beatriz Moraes

.....

